Examiner only

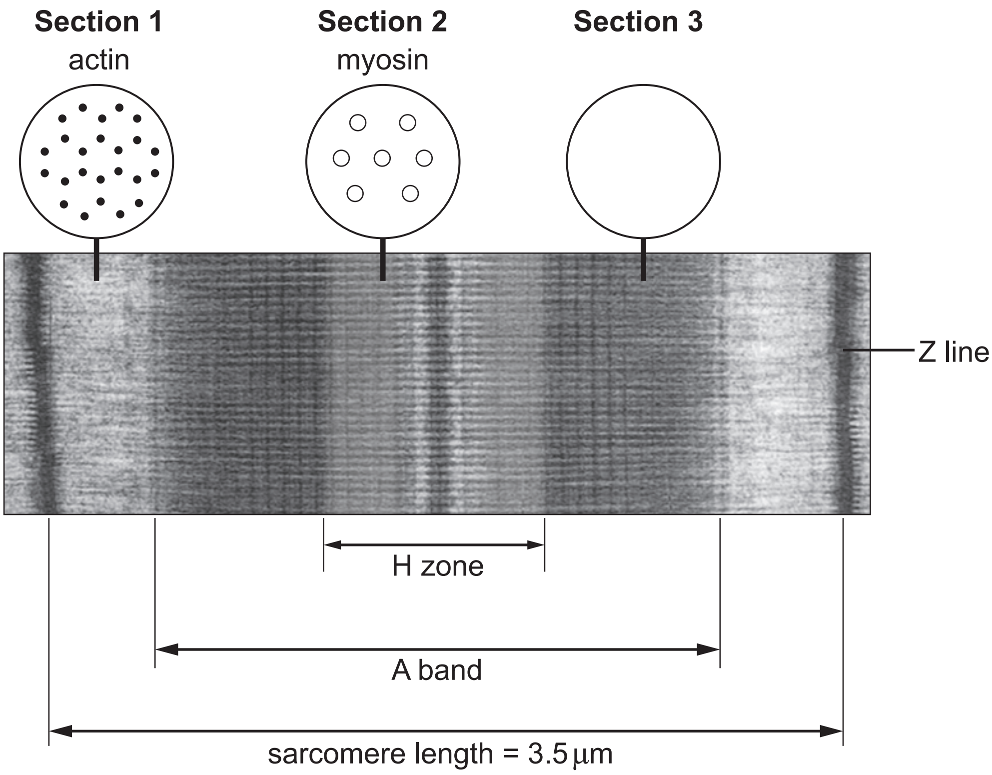
Option B: Human Musculoskeletal Anatomy

8. The human musculoskeletal system is comprised of a number of different tissues, including muscle, cartilage and bone.

Image 8.1 shows the structure and appearance of a sarcomere in relaxed, striated muscle. It also shows transverse sections of the actin and myosin filaments in the sarcomere.

Image 8.1

Transverse sections across sarcomere



(a) (i) **Complete the appearance** of transverse **Section 3** shown on **Image 8.1**. [1]

(ii) During contraction of the muscle, the sarcomere shortens. Describe and explain what happens to the H zone and A band. [4]

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(iii) When the sarcomere is contracted, its length is 1.5 μm long.
Use **Image 8.1** to calculate the percentage decrease in length when the sarcomere contracts. [2]

Percentage decrease in length of sarcomere = ...

(iv) Describe how the structure of actin allows cross bridges to form during contraction of the muscle and explain why cross bridges are unable to form in a resting muscle. [2]

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(b) Cartilage is connective tissue found in the skeleton. There are several forms of cartilage.

(i) Explain how the structure of hyaline cartilage relates to its function in the skeleton. [2]

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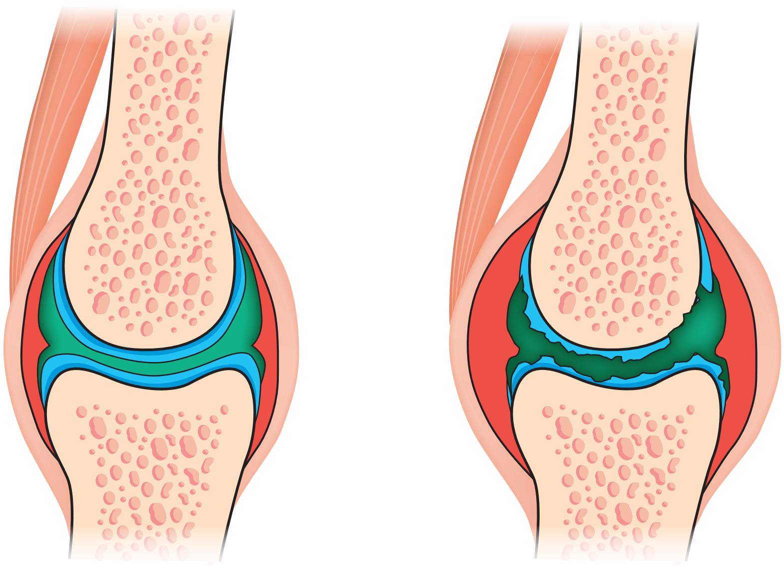
Image 8.2 shows a normal knee joint and a joint affected by rheumatoid arthritis.

Image 8.2



normal knee joint

knee joint with rheumatoid arthritis

(ii) Describe what is meant by rheumatoid arthritis and name **two** types of tissues affected by rheumatoid arthritis as shown in **Image 8.2**. [2]

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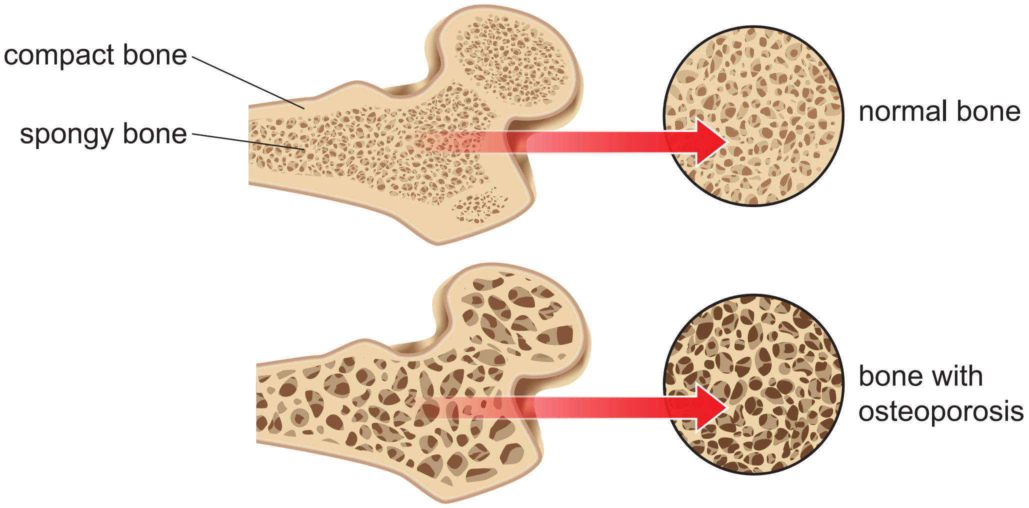
(c) Osteoporosis is a form of brittle bone disease in humans. It is caused by the loss of calcium from the bones leading to decreased bone density. **Image 8.3** shows a cross section of bone from a healthy patient, and one from a patient suffering from osteoporosis.

Image 8.3



(i) Using **Image 8.3** and your own knowledge, describe the effect of osteoporosis on the composition of the bones. [2]

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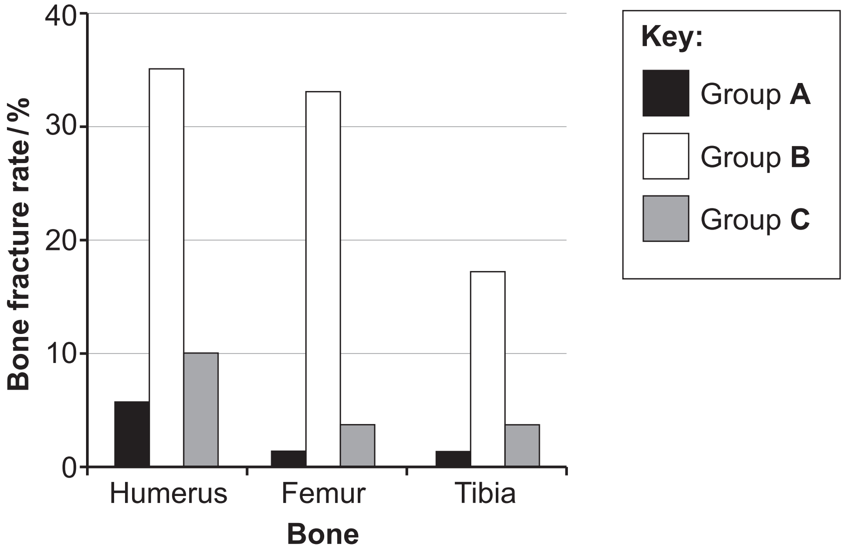
Amniotic fluid contains a form of stem cell that can differentiate into osteoblasts. Research has been carried out to assess the effectiveness of stem cells in treating osteogenesis imperfecta (OI) in mice.
Three groups of mice were tested. The treatment received by each group is shown in **Table 8.4**.

Table 8.4

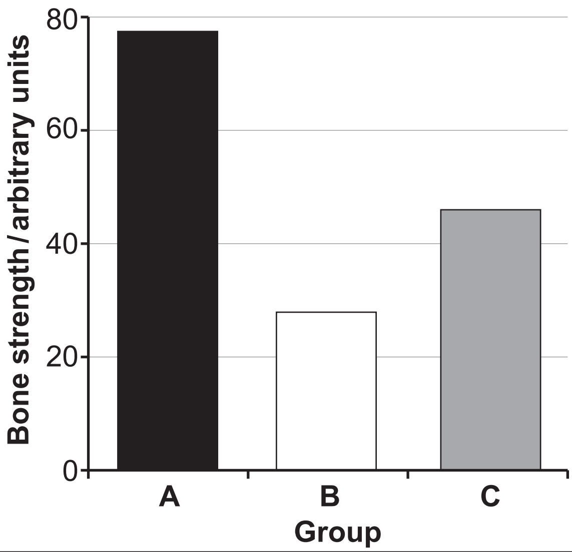
Group of mice	Osteogenesis imperfecta (OI) present	Received amniotic stem cells at birth
A – healthy mice	No	No
B – control group	Yes	No
C – experimental group	Yes	Yes

The results of the research are shown in **Graphs 8.5** and **8.6**.

Graph 8.5



Graph 8.6



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(ii) Summarise the effect of stem cell treatment in the experimental group **C** in comparison to the healthy mice (**A**) and the control group (**B**) shown in **Graphs 8.5 and 8.6**. [3]

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(iii) The use of human stem cells is being considered for clinical trials in humans to treat osteoporosis. Using all of the information provided, suggest how stem cell treatment could improve the condition of bones in osteoporosis patients. [2]

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